

RESKO-SMOLSKO, Poland

WMO: 122100

Lat: 53.7636N

Lon: 15.3933E

Elev: 173

StdP: 14.60

Time Zone: 1.00 (EUC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.5	15.1	3.9	6.8	12.4	9.1	8.8	17.6	21.4	41.5	19.2	41.2	3.1	70	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	84.5	67.2	81.1	65.8	78.0	64.7	69.5	80.2	67.8	77.8	66.2	74.7	6.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	95.5	73.1	64.3	90.9	71.7	62.9	86.3	70.2	33.6	80.6	32.3	77.5	31.0	74.4	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.5	12.7	DB	2.8	91.3	9.1	3.8	-3.7	94.1	-9.1	96.3	-14.1	98.5	-20.7	101.3	(4)
(5)			WB	2.4	72.9	8.7	2.4	-3.8	74.6	-8.9	76.0	-13.8	77.3	-20.1	79.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	30.9	32.9	38.3	47.6	54.8	60.7	64.6	64.5	57.0	48.9	41.3	35.3		
	DBStd	13.57	8.85	7.83	7.76	7.00	7.00	5.91	5.34	5.26	5.39	6.34	5.88	7.85		
	HDD50	2416	591	479	366	127	28	1	0	0	6	98	264	456		
	HDD65	6320	1056	899	828	524	327	159	75	74	247	499	711	921		
	CDD50	1741	0	0	4	54	176	321	452	450	216	64	4	0		
	CDD65	167	0	0	0	0	10	29	62	59	7	0	0	0		
	CDH74	1711	0	0	0	11	128	336	654	516	66	0	0	0		
	CDH80	422	0	0	0	2	20	88	173	131	8	0	0	0		
(14) Wind	WSAvg	5.8	6.6	6.4	6.4	5.8	5.5	5.4	5.1	5.1	5.2	5.5	6.1	7.0		
(15) Precipitation	PrecAvg	27.6	2.1	1.7	2.0	1.4	2.1	2.8	3.1	3.2	2.5	2.0	2.2	2.4		
	PrecMax	36.9	5.0	3.3	3.8	4.1	3.5	7.0	6.7	6.7	5.8	4.9	4.8	4.3		
	PrecMin	22.8	0.0	0.3	0.4	0.4	0.4	1.1	0.1	0.7	0.7	0.9	0.1	0.6		
	PrecStd	3.8	1.2	0.8	0.9	0.8	0.8	1.1	1.7	1.4	1.1	1.1	1.0	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.1	52.7	63.2	74.6	82.8	88.0	89.4	89.0	81.2	69.4	57.3	53.3		
		MCWB	47.7	47.5	50.8	58.2	64.9	69.0	69.4	69.3	65.8	59.5	54.4	50.5		
	2%	DB	47.2	48.9	57.2	69.1	77.7	82.0	84.6	83.3	75.4	65.1	53.7	49.9		
		MCWB	44.6	44.9	47.6	54.6	62.1	66.1	67.5	66.7	62.0	57.9	50.6	47.6		
	5%	DB	44.7	46.1	52.9	65.1	73.1	77.6	80.9	79.6	71.5	61.6	51.6	47.5		
		MCWB	42.7	42.8	46.8	52.6	59.5	64.8	66.3	65.1	60.7	56.3	48.9	45.2		
	10%	DB	42.3	43.4	49.5	61.2	69.0	73.6	77.3	75.9	67.4	58.6	49.6	44.9		
		MCWB	40.4	40.5	44.5	50.8	57.7	62.7	65.0	64.3	59.0	54.5	47.5	42.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.5	49.0	52.5	60.3	67.3	72.2	71.4	71.4	67.7	61.3	55.4	51.1		
		MCDB	51.1	51.5	59.3	72.3	78.4	83.9	83.8	83.7	77.0	67.0	57.2	52.8		
	2%	WB	44.9	45.8	49.7	55.9	63.9	68.0	69.4	69.1	63.7	58.9	51.2	48.1		
		MCDB	46.7	48.2	54.8	66.5	73.8	79.5	80.7	79.1	72.0	63.5	53.1	49.7		
	5%	WB	42.9	43.0	47.0	53.7	61.5	66.0	67.9	67.3	61.7	56.8	49.3	45.6		
		MCDB	44.5	45.5	51.8	63.0	70.4	75.1	78.1	75.8	68.6	60.8	51.1	47.3		
	10%	WB	40.7	40.5	44.9	51.4	58.9	63.8	66.1	65.6	60.0	54.9	47.6	43.1		
		MCDB	42.4	43.2	49.1	59.7	66.8	71.6	74.6	72.7	66.4	58.3	49.4	44.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	9.6	13.5	18.2	19.6	19.0	18.3	18.4	16.5	12.8	8.1	6.9		
		MCDBR	7.6	11.8	19.5	25.4	25.7	26.0	25.9	25.2	23.0	17.4	9.9	8.5		
		MCWBR	6.9	8.9	11.9	13.8	12.9	13.4	11.2	11.3	11.5	11.2	7.7	7.8		
	5% WB	MCDBR	7.6	10.4	16.7	22.9	22.8	23.2	22.5	20.7	19.9	16.0	9.3	8.5		
		MCWBR	7.1	8.7	11.1	13.0	12.4	12.9	10.7	10.5	11.2	10.9	7.8	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.312	0.337	0.369	0.407	0.405	0.401	0.424	0.421	0.381	0.361	0.339	0.309		
	taud		2.301	2.299	2.265	2.224	2.279	2.317	2.268	2.286	2.376	2.391	2.357	2.307		
	Ebn at Noon		204	238	255	260	267	269	260	253	249	227	192	180		
	Edh at Noon		19	26	33	39	39	38	40	37	30	24	18	16		
(44) All-Sky Solar Radiation	RadAvg		172	380	768	1324	1666	1760	1627	1370	995	514	210	120		
	RadStd		18	55	86	175	194	136	206	119	107	75	29	11		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-252	+108	N/A

Nomenclature: See separate page